Mahavir Education Trust's

Shah & Anchor Kutchhi Engineering College

Chembur, Mumbai-400088

(An Autonomous Institute Affiliated to University of Mumbai)



Department of Cyber Security

presentation
on

My Self-Learning Journey

(Subject:Advanced Digital Forensics HNCSCR0701)

by

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Vision & Mission

Institute Vision

To become a globally recognized institution offering quality education and enhancing professional standards

Department Vision

To be a high repute focused department preparing students ethically in field of cyber security and make them capable to lead technical, economic, social and ethical development of the society.

Institute Mission

To impart high quality technical education to the students by providing an excellent academic environment, well - equipped laboratories and training through motivated teachers.

Department Mission

1. To provide an academic environment for the development by providing training and skill development to build knowledge, skills, values, and confidence to take a leadership role in the field of Cyber Security.
2. To cultivate research and entrepreneurship culture resulting in knowledge and innovative technologies that contribute to sustainable development of the society.
3. To enhance academic and corporate collaborations.

Program Specific Outcomes

Index	Outcome
PSO1	The Cyber Security graduates are able to gain a thorough understanding of the Cyber Security landscape with its growing threats and vulnerabilities in the world of computing including software and hardware.
PSO2	The Cyber Security graduates are able to attain skills to comprehend and anticipate future challenges and devise methods to meet them and also, be articulate and skilled to convince all the stakeholders
PSO3	The Cyber Security graduates are able to acquire and demonstrate the ability to use ethical standard tools, practices and technologies for the analysis, design, development, implementation and testing of innovative and optimal Cyber Security solutions without compromising the privacy needs of individual and entities and the security concerns of law enforcement agencies.

Outline

- Introduction
- Problem Statement & Objectives
- Core Concept / Theory
- Working / Mechanism
- Functional & Non-Functional Requirements (if applicable)
- Challenges / Limitations
- Case Study / Example
- Comparative Analysis (if applicable)
- Tools & Modeling Approaches
- Applications in Real Life
- Proposed Framework / Design
- Conclusion & Future Scope
- Reference

My Self-Learning Journey

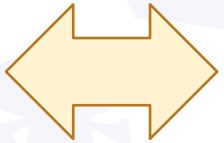
Purpose: to reflect on what I learned from two self-paced cybersecurity courses, how I worked through them, and where they fit in my plans.

Cisco Ethical Hacker

Cisco Networking Academy

The wide view: how an authorised security assessment is scoped, carried out and reported.

Certificate date: 16 Sep 2026



Kali Linux Advanced Wireless Penetration Testing

Infosys Springboard

The close-up: how the security of wireless networks is assessed.

Completed: September 2, 2026

Independent learning

Working through self-paced material on my own

Technical understanding

Connecting ideas across the two courses

Career development

Building a base for security-related roles

Activity 1: Cisco Ethical Hacker

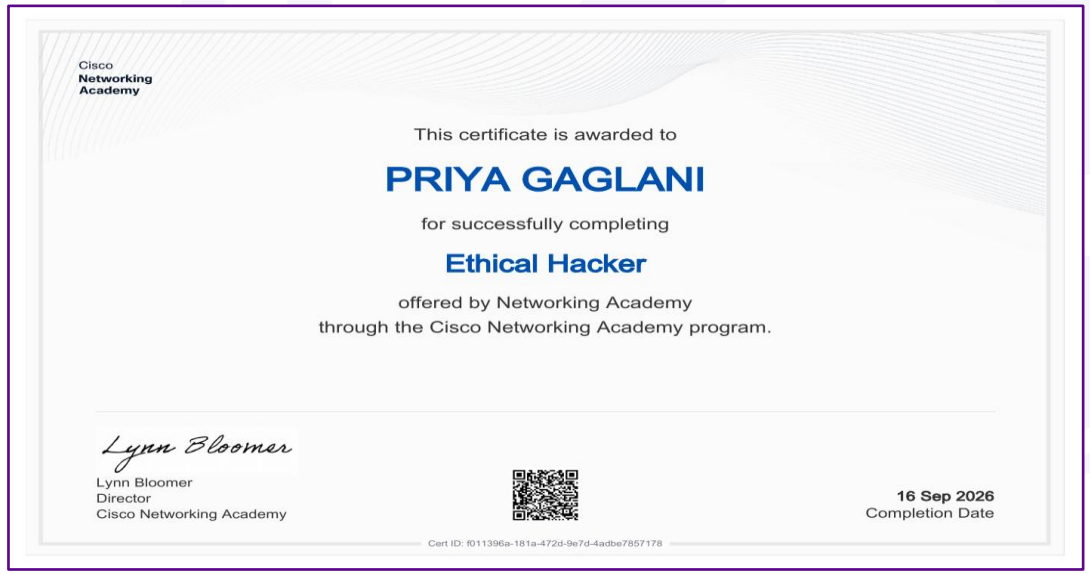
Course snapshot

Provider: Cisco Networking Academy

Course: Ethical Hacker

Certificate date: 16 Sep 2026

Completion: Final assessment passed; badge awarded


Cisco Networking Academy

This certificate is awarded to

PRIYA GAGLANI

for successfully completing

Ethical Hacker

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer
Director
Cisco Networking Academy

16 Sep 2026
Completion Date

Cert ID: f011396a-181a-472d-9e7d-4adbe7857178

1

Why it matters

Ethical hackers test systems with permission, so weaknesses are found and fixed before an attacker finds them.

2

How the course is built

Ten modules, starting with ethical hacking and penetration testing basics, then planning and scoping an assessment.

3

What I took from it

Cisco describes the course as scoping, executing and reporting. For me the useful part was seeing the whole assessment as a process, not a toolbox.

Activity 2: Wireless Penetration Testing

Course snapshot

Provider: Infosys Springboard

Course: Kali Linux Advanced Wireless Penetration Testing

Completed: September 2, 2026 (issued Sep 3, 2026)

Verification: QR code on the certificate



Infosys
Navigate your next

COURSE COMPLETION CERTIFICATE

The certificate is awarded to

Priya Gaglani

for successfully completing the course

Kali Linux Advanced Wireless Penetration Testing

on September 2, 2026

Infosys | Springboard

Congratulations! You make us proud!

Satheesha B. N.
Senior Vice President and Head
Education, Training and Assessment
Infosys Limited

Issued on: Thursday, September 3, 2026
To verify, scan the QR code at https://verify.onwingspan.com

Ethical hacking (Cisco)

the whole assessment process: scope, assess, report

Wireless security (Infosys)

one area looked at in depth: Wi-Fi networks and how they are protected

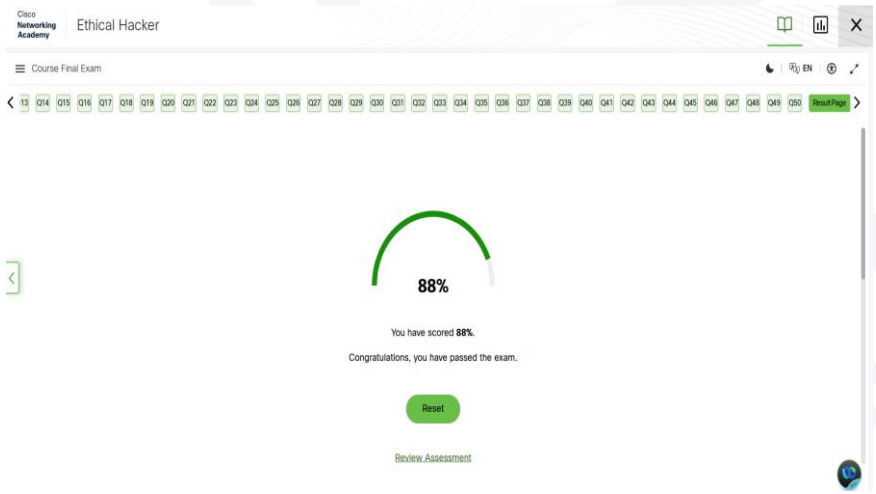
How I see the link

- Wireless networks are how most devices connect, so they are a natural area to assess.
- Kali Linux is a Linux distribution that bundles security tools; the course uses it as its setting.
- I claim the course and the concepts, not lab experience. Hands-on practice is a next step.

Learning Process and Evidence

1 Study	2 Flag	3 Revisit	4 Check
Work through the course material module by module	Mark terms and ideas I do not recognise	Return to harder topics and re-explain them simply	Use the assessments to test what I actually understood

Resources: the Cisco Networking Academy and Infosys Springboard course content. The Cisco final assessment was part of the completion process.



Evidence 1: Cisco final assessment score



Evidence 2: Cisco course badge



Evidence 3: Infosys Springboard certificate

Key Technical Concepts

1 Scope and permission → 2 Reconnaissance → 3 Assessment → 4 Document and fix

Conceptual illustration of an authorised assessment. It is not a record of tests I carried out myself.

	Ethical hacking (Cisco)	Wireless testing (Infosys)
Scope	Whole systems and networks	Wi-Fi networks and access points
Core question	Where could an attacker get in?	How well is the wireless link protected?
Typical output	Scoped assessment and a report	Wireless-specific findings and fixes

Role of Kali Linux

A Linux distribution that comes with many security-testing tools, so a tester has a ready environment. It is only used on systems the tester owns or has written permission to test.

Practical Example (Hypothetical)

HYPOTHETICAL EXAMPLE illustrative only, not an exercise I performed

Scenario: a small cafe asks a tester for a written review of its own Wi-Fi setup.

1 Permission and scope

Signed approval from the owner. Only the cafe's router and its guest network are in scope.

2 Identify a concern

Staff and guest devices share one network, secured by a short passphrase that has never been changed.

3 Examine evidence

Review the router settings with the owner: encryption mode, firmware version, network separation.

4 Recommend a fix

Separate guest network, strong unique passphrase, firmware update, WPS off, then re-check and document.

Safe by design: the example stops at configuration review and recommendations. No attack steps, no capturing of anyone's credentials.

Skills Gained

Technical knowledge

- The scope, assess and report flow of an ethical hacking engagement
- Why wireless networks need their own testing attention
- What Kali Linux is and why testers use it
- Security vocabulary I can now follow without constant look-ups

Analytical thinking

Asking what could go wrong, then checking the evidence

Self-learning

Planning my own study and finishing two courses

Communication

Explaining technical ideas plainly in this reflection and video

Documentation

Stating a finding, its risk and the fix clearly

Ethical responsibility

Permission and scope come before any testing

Professional growth

Verified certificates and concrete points to discuss

Course completion shows foundation-level understanding. It is not professional expertise, and I treat it that way.

Challenges and How I Addressed Them

Learning challenge	Learning approach
Unfamiliar terminology	Look each term up, then explain it back in my own words
Linking two courses	Map the shared stages (scope, assess, report) onto the wireless topics
Knowing if I really understood	Use assessments, including the Cisco final assessment, to spot gaps and revisit them
Learning offensive ideas responsibly	Keep authorisation in view; practise only in authorised labs

Career Relevance

Computer Engineering base
Programming, networks, systems

Two security courses
Ethical Hacker + wireless testing

Relevant starting points
Entry-level security, security testing, software roles where secure design matters

Placement preparation

- Interview vocabulary I can explain
- Two verified certificates to discuss

Project work

- Think about misuse and weak points early
- Write clearer notes on security decisions

Higher studies

- A base for network or wireless security study
- Helps me choose a research direction

Certificates strengthen my profile. They do not guarantee a job.

Reflection and Future Learning Plan

My main takeaway

Ethical hacking starts with permission and scope, and tools come after that.

Doing two courses showed me the same principle from a wide view and a wireless view, and it changed how I think about “testing”.

1. **Strengthen networking fundamentals**
   TCP/IP and Wi-Fi standards, so concepts connect faster
2. **Practise only in authorised labs**
   My own test setup or a college lab, with permission
3. **Improve documentation**
   Write sample assessment reports with finding, risk and fix
4. **Keep learning in a structured way**
   Pick the next course after reviewing gaps from these two

Evidence and Overall Takeaways


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Certificate 1: Cisco Ethical Hacker, 16 Sep 2026


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Satheesha B. Nanjappa
Senior Vice President and Head
Education, Training and Assessment
Infosys Limited

Certificate 2: Infosys Springboard, Sep 2, 2026

Completed

Two courses from two providers, each with a certificate

Learned

Assessment is a process: scope, assess, report, and improve

Next

Networking basics & authorised labs

Thank You